Individual-tree QSM reconstruction and above-ground biomass agreement on TreeScanPL10k: a methodological note for TWIG

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Abstract

I applied the Arbor QSM pipeline to 30 TreeScanPL10k plots (1,616 manually labeled trees) to test whether QSM-derived tree metrics and AGB are reliable enough for multi-temporal TLS monitoring. The mesh pipeline fails catastrophically on DBH on most plots (100% failure on 22 of 30), with fitted diameters of 600–2000 cm (10–100× the true diameter), and this failure is not associated with the stand's height IQR (r = −0.05, R^2 = 0.00). The QSM pipeline's DBH failure is lower and is weakly associated with IQR (r = 0.31, R^2 = 0.10). QSM AGB, computed as total wood volume × wood density, is associated with Chave allometric AGB at R^2 = 0.453 (N = 1,581). Because the Chave model is pantropical and these are temperate stands, the allometric reference is likely the noisier of the two. This R^2 is a lower bound on QSM accuracy for trees that QSM reconstructs, not an absolute calibration. QSM wood volume is also strongly rank-correlated with a raw-point voxel-volume proxy (R^2 = 0.615, N = 1,614), which shows that QSM preserves tree-to-tree ordering of structural size but is not itself a biomass comparison. The height IQR diagnostic is weakly informative for QSM DBH and not informative for tree detection, height, or per-plot AGB agreement. Per-plot AGB agreement is bounded by QSM fitting noise and allometric prediction error rather than by IQR. Of the two representations compared, QSM's DBH failures are lower on average and are weakly associated with IQR (r = 0.31), while the mesh's DBH failures occur on almost every plot regardless of IQR (r = −0.05).

Introduction

TWIG will convert repeated TLS acquisitions into multi-temporal AGB estimates at Amacayacu and Alto da Figueira. The workflow requires two things that are not currently well-characterized in the TLS literature: (i) a tree segmentation and reconstruction method that does not catastrophically fail on any structural class, and (ii) a pre-processing diagnostic that tells the analyst which plots will produce usable metrics before time is spent on full processing. This note reports the results of a systematic comparison of two representations — a mesh-based pipeline (*TopTreeSegR*, https://github.com/DijoG/TopTreeSegR) and a QSM pipeline (*Arbor*, https://www.r-lidar.com/arbor) — applied to the 30 TreeScanPL10k plots (Stereńczak et al., 2026). The goal is not to rank the representations, but to characterize their failure modes and to identify the stand-level diagnostic that is associated with them. The findings are directly relevant to TWIG's multi-temporal design.

Data and methods

Data

TreeScanPL10k provides 30 manually labeled TLS plots across six Polish forest districts, spanning pine, spruce-pine, mixed, and broadleaf stands. Each plot contains 10–157 individually labeled trees (1,616 total), with reference DBH (Taubin circle fit in the 1.15–1.45 m band) (Taubin, 1991), height (z-range), and crown volume (0.3 m voxel occupancy above 60%

of tree height). The plots span a vertical structural heterogeneity range from 1.3 m to 17.5 m in reference-height IQR.

QSM pipeline

Arbor was applied to each plot with default parameters. The pipeline segments the point cloud semantically (ground, wood, foliage) and fits cylinder models to each tree, producing per-tree DBH (from the taper fit), height (from the uppermost cylinder), and total wood volume V (sum of cylinder volumes from base to tip). Crown volume was computed as the sum of cylinder volumes above 60% of tree height, but is not used in the AGB comparison because it measures the cylinder envelope of the crown (wood plus a small amount of air), which is not comparable to the voxel-occupancy crown volume used for the mesh pipeline.

Mesh pipeline

TopTreeSegR was applied to the same semantically filtered point clouds, so that both representations were evaluated on identical input. The pipeline constructs an α-complex mesh (α = 0.05 m), computes a Forman gradient of the height function, identifies density-based seed minima, and traces ascending manifolds to partition the canopy into individual trees. DBH is fitted by a circle in the 1.15–1.45 m band, height is the z-range of the assigned points, and crown volume is computed by 0.3 m voxel occupancy (Calders et al., 2015) above 60% of tree height.

AGB calculation

Three quantities were computed per tree: (i) **QSM AGB** = V × ρ × 1000, where V is the total wood volume in m³ and p is the species-level or forest-type-level wood density in $\mathrm{g/cm^3}$; (ii) **raw-point voxel volume** = V_voxel, the occupied volume of all tree points for the whole tree, voxelized on a 0.3 m grid. This is an envelope volume that includes air and foliage between branches and is not an AGB estimate; it is used only as a structural proxy to test whether QSM preserves the tree-to-tree ordering of size. (iii) **allometric AGB** = 0.0673 × (ρ × $\mathrm{m^2}$ × H)^0.976 (Chave et al., 2014), applied to the reference DBH and height. Wood density (ρ) was assigned by forest type: 0.45 for Gorlice mixed, 0.45 for Herby pine, 0.42 for Piensk pine, 0.42 for Katrynka and Suprasl spruce-pine, 0.50 for Milicz mixed. All AGB values are reported in tonnes (Mg).

QSM AGB, as computed here (wood volume × wood density), is a relative measure of woody biomass: it preserves the tree-to-tree ordering of size and is usable for multi-temporal change detection and for flagging reconstruction failures, but it is not an absolute biomass measurement. Chave et al. (2014) is a pantropical model and is not species- or region-specific for these temperate Polish stands, and wood density was assigned at the forest-type level. The allometric reference therefore carries its own systematic error and is likely the noisier of the two estimates. Any $\mathrm{R^2}$ between QSM and allometric AGB is consequently a lower bound on QSM accuracy for trees that QSM reconstructs, rather than an estimate of absolute QSM error. Converting QSM AGB to absolute tonnes would require either a local destructive sample or a species-specific allometric equation calibrated against destructive data.

Of the 1,616 labeled trees, 1,614 have both a valid QSM wood volume and a valid raw-point voxel volume; of those, 1,581 also have a valid reference DBH and height and therefore a valid allometric AGB. The losses are due to trees with no valid cylinder fit in the QSM output and to trees with no measurable reference DBH, and are not expected to bias the comparison.

Failure flags

A tree's metric was flagged as a catastrophic failure when the ratio of predicted to reference value fell outside [0.5, 2.0] for DBH and height, or outside [0.4, 2.5] for crown volume. These bounds correspond to a predicted value that is less than half or more than double the reference value, which is the range in which the tree's metric would be unusable for biomass estimation; the bounds are deliberately wider than the 5–10% relative error typical of TLS DBH measurement, so that the flag isolates gross segmentation or fit failures rather than ordinary measurement noise. A tree was flagged as a detection failure when it was present in the reference but absent from the pipeline output. The stand-level predictor was the interquartile range (IQR) of reference tree heights, computed before any TLS processing: the IQR measures the vertical spread of the canopy (the difference between the 75th and 25th percentile of tree heights), and is a single number per plot that a forester can estimate from a coarse pre-segmentation crown delineation.

Results

Tree metrics

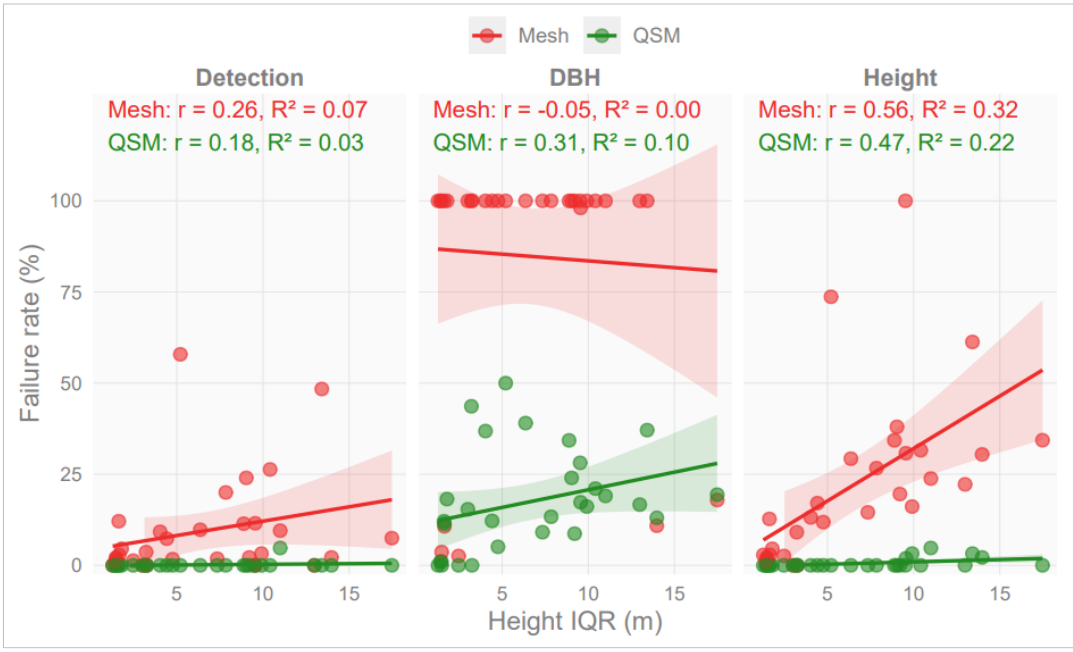


Figure 1. Detection failure, DBH failure, and height failure rates as a function of stand-level height IQR. Points are per-plot failure rates (%; 30 plots), colors are the two representations, lines are linear fits with 95% confidence bands, and the r and R^2 in each panel are computed per method. The mesh's DBH failure is at 100% on most plots and is not associated with IQR (r = -0.05, R^2 = 0.00); on Gorlice_2015_0101703, for example, the circle fit produces fitted DBHs of 600–2000 cm, 10–100× the true diameter (tree-level output, not shown). The QSM's DBH failure is lower and is weakly associated with IQR (r = 0.31, R^2 = 0.10). In the Detection panel, the mesh's failure is higher and is weakly associated with IQR (r = 0.26, R^2 = 0.07), while the QSM's failure is low and is not associated with IQR (r = 0.18, R^2 = 0.03). In the Height panel, the mesh's failure is higher and is moderately associated with IQR (r = 0.56, R^2 = 0.32), while the QSM's failure is low and is weakly associated with IQR (r = 0.47, R^2 = 0.22).

AGB comparison

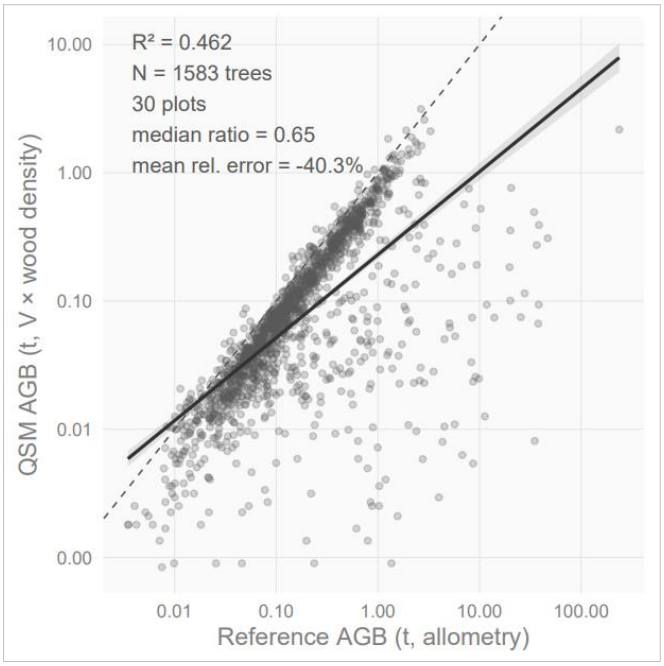


Figure 2. QSM AGB (total wood volume × forest-type wood density) against Chave allometric AGB, for the 1,581 of 1,616 trees with both a usable QSM volume and a reference DBH and height (30 plots). Both axes are tonnes (Mg) on $\log_{10}$ scale; the dashed line is 1:1 and the solid line is the $\log_{10}$-$\log_{10}$ least-squares fit (R^2 = 0.453, slope = 0.65, intercept = −0.65). The fit lies below 1:1 across the range (median QSM/allometry ratio 0.64; mean relative error −40.3%), because QSM reconstructs only the woody skeleton and excludes fine branches and bark, whereas the allometric equation estimates total above-ground biomass. The mean is pulled down by a tail of trees where QSM reconstructs only the stem. The agreement on log axes shows that QSM preserves the tree-to-tree ordering of biomass, which is the property required for multi-temporal change detection; the absolute scale is not calibrated by this comparison.

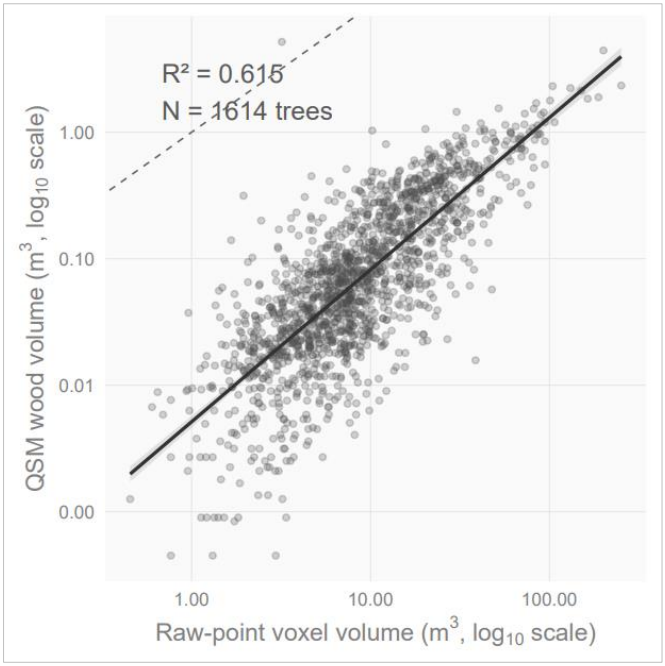


Figure 3. QSM wood volume against raw-point voxel volume (0.3 m occupancy), 1,614 trees. Both axes are volumes in m^3. QSM measures the woody skeleton; voxel occupancy measures

the occupied crown envelope, including air between branches and foliage. The two are not on the same scale, and the factor-100 offset is the wood-to-envelope volume ratio, which depends on crown openness and varies by species and structure. The R^2 = 0.615 on log axes shows that QSM preserves the tree-to-tree ordering of structural size, which is the property that matters for multi-temporal change detection; it is not evidence that QSM AGB is accurate in absolute terms (see Fig. 2 for the allometric comparison).

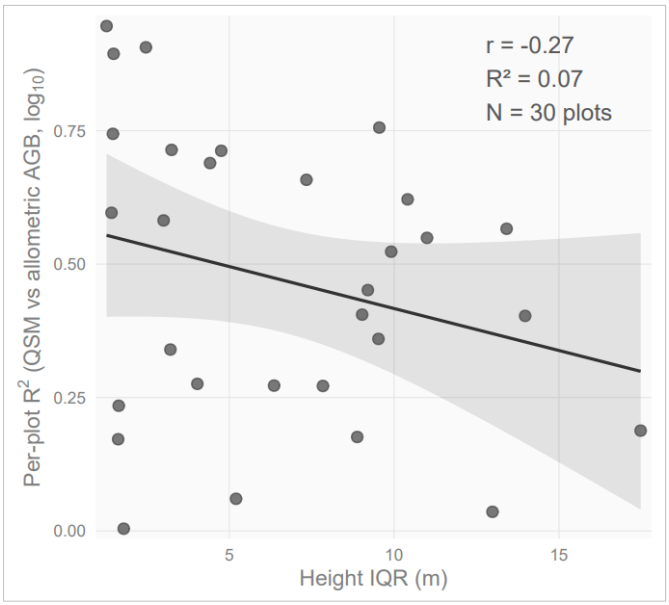


Figure 4. Per-plot QSM-vs-allometric AGB agreement ($\log_{10}$-$\log_{10}$ R^2) against the reference-tree height IQR, one point per plot (N = 30). The line is an ordinary least-squares fit with a 95% confidence band. The association is weak and slightly negative (r = −0.27, R^2 = 0.07): per-plot AGB agreement is not predicted by the stand-level height diagnostic that is associated with QSM DBH failure (Fig. 1). The sign is opposite to the direction that would indicate worse agreement in structurally complex plots, and the fit explains 7% of the variance, so we interpret this as no association.

Discussion and conclusions

The introduction asked two questions that the TLS literature does not yet answer well: whether a tree reconstruction method exists that does not catastrophically fail on any structural class, and whether a stand-level diagnostic can predict which plots will produce usable metrics before full processing. The results give clear answers to both.

On catastrophic failure. Of the two representations compared, the mesh pipeline fails catastrophically on DBH on almost every plot: on 22 of 30 plots, the failure rate is 100%, and the fitted diameters are 600–2000 cm (10–100× the true value) because the circle fit in the 1.15–1.45 m band captures the union of adjacent stems (Fig. 1). The QSM pipeline does not have this failure mode: its taper fit starts at the base and grows upward, separating adjacent stems by construction. The QSM's DBH failure is at 0–50% on the low-IQR (< 10 m) plots and below 25% (except one) on the high-IQR plots. The strongest single finding of the note is therefore not that QSM is universally reliable, but that it has lower catastrophic failure than the mesh and that its failure rate is weakly associated with the stand-level diagnostic, whereas the mesh fails on almost every plot regardless of IQR.

On the stand-level diagnostic. The IQR of reference tree heights is associated with QSM DBH failure (r = 0.31, R^2 = 0.10) but not with mesh DBH failure (r = −0.05, R^2 = 0.00). This is the panel where the diagnostic is most useful for QSM and least useful for the mesh. In the Detection panel, both methods show a weak positive association (Mesh r = 0.26, QSM r = 0.18). In the Height panel, the mesh is more strongly associated with IQR than the QSM (Mesh r = 0.56, QSM r = 0.47), which is the opposite of the DBH panel. Per-plot AGB agreement is not associated with height IQR (r = −0.27, R^2 = 0.07, N = 30; Fig. 4). The diagnostic that predicts QSM DBH failure therefore does not predict QSM AGB agreement: high-IQR plots can still have usable AGB agreement even when individual DBH measurements fail, and low-IQR plots can have poor AGB agreement for reasons unrelated to stand structure. This answers the second question with a qualification: the height IQR is weakly informative for QSM DBH, not informative for detection or height, and not informative for AGB agreement.

Implications for TWIG. For the multi-temporal AGB workflow, three operational conclusions follow. First, QSM should be the reconstruction method for high-IQR plots, where the mesh fails unconditionally, because QSM's catastrophic DBH failures are lower than the mesh's and are weakly associated with IQR. Second, the height IQR is a weak pre-processing indicator, informative enough to warn that high-IQR plots are more likely to produce QSM DBH failures (r = 0.31, R^2 = 0.10), but not informative enough to justify excluding plots. The mesh pipeline has no such pre-processing check, because it fails on almost all plots regardless of IQR. Third, QSM AGB tracks Chave allometric AGB across the tree-size range (R^2 = 0.453, a lower bound on QSM accuracy; see Methods), with a systematic median underestimation of 0.64 of the allometric value (mean relative error −40.3%; Fig. 2). This offset is consistent with QSM reconstructing only the woody skeleton and not foliage or fine branches. QSM wood volume also preserves the tree-to-tree ordering of structural size relative to a raw-point voxel-volume proxy (R^2 = 0.615), so it is usable as a relative-change estimator. Absolute AGB calibration can be established from allometry or from a small destructive sample, and the residual variance is an open question that TWIG's multi-temporal design is positioned to resolve. The height IQR is one instance of a broader class of pre-processing proxies: the framework that produced it — tree-level failure flags aggregated per plot — can be applied to any stand-level variable, such as canopy cover, height heterogeneity, point-density profile, or return intensity, to identify the proxy that best predicts reconstruction failure in a given forest type. Return intensity is the most promising of these, because recent work links it to wood recall and structural connectivity in QSM reconstruction (Kaharlytskyi et al., 2026).

Supplementary figures

Figure S1. Per-plot QSM AGB against Chave allometric AGB, one panel per plot (N = 30), ordered by per-plot $\log_{10}$-$\log_{10}$ R^2 in descending order. Strip labels report the plot identifier (site and plot number), the per-plot R^2, the median QSM/allometry ratio, and the number of matched trees. Both axes are on $\log_{10}$ scale and free within each panel so that small plots remain legible. Dashed line is 1:1. Per-plot R^2 ranges from 0.00 to 0.95, and the median QSM/allometry ratio ranges from 0.23 to 0.80. The pooled relationship across all plots is shown in Fig. 2 of the main text.

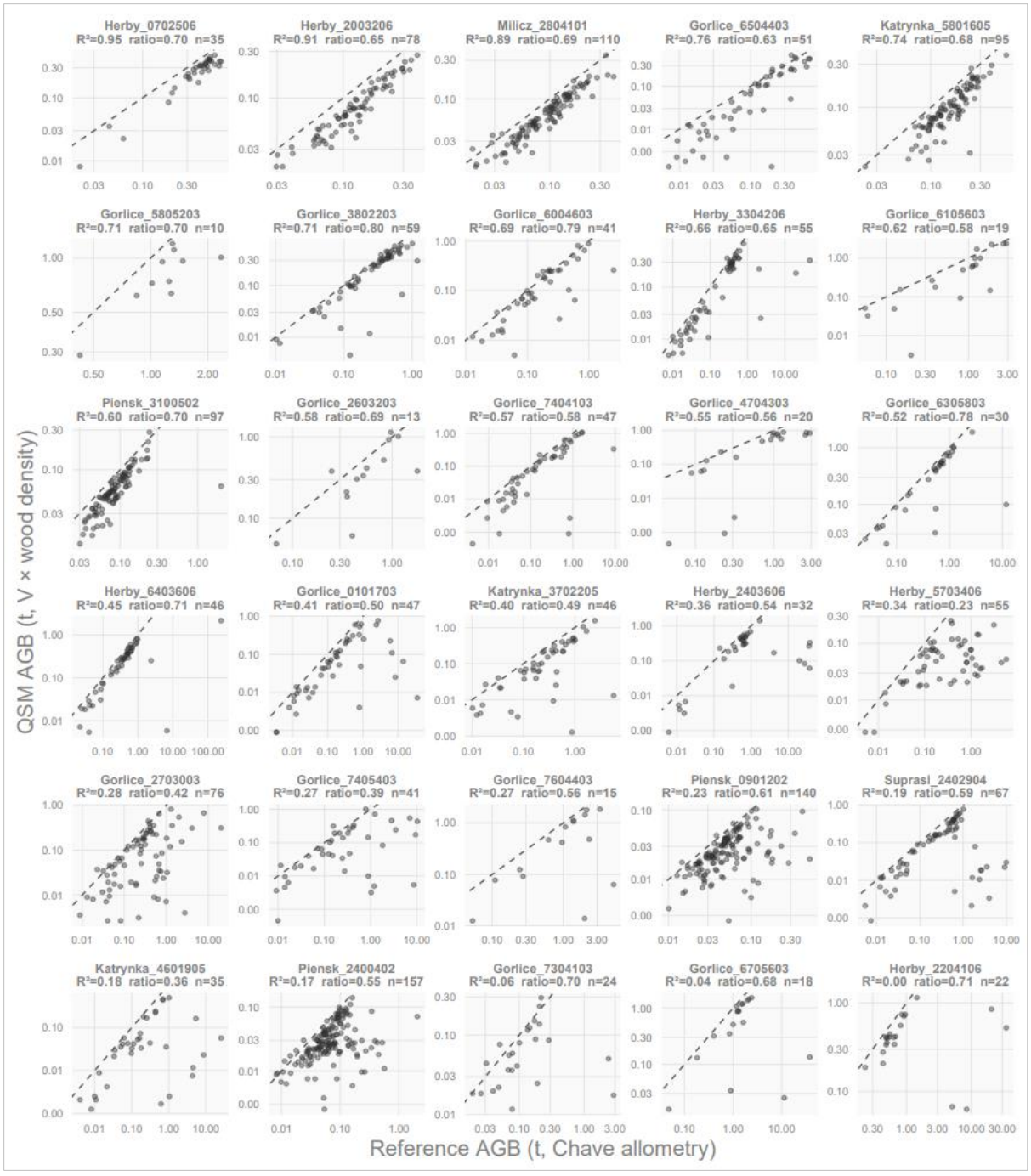
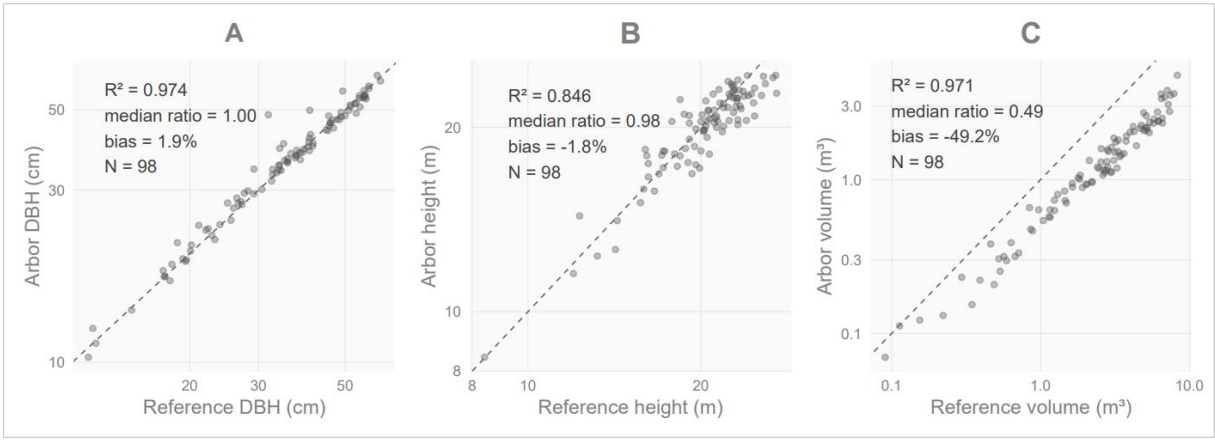


Figure S2. Arbor QSM estimates against the destructive reference for 98 trees from Québec, Canada (MLS, hardwood; sugar maple and yellow birch). (A) DBH, (B) standing height, (C) total woody volume. Both axes are on $\log_{10}$ scale; the dashed line is 1:1. For each panel, R^2 and slope are from a $\log_{10}$-$\log_{10}$ least-squares regression of Arbor on the reference; "median ratio" is the median of Arbor/reference across trees; “bias” is the mean of (Arbor − reference)/reference × 100. Reference height is the standing height where available, and felled length plus stump height where not. Arbor reproduces DBH almost exactly (bias +1.9%, R^2 = 0.974, median ratio 1.00) and standing height with a small negative bias (−1.8%, R^2 = 0.846, median ratio 0.98), and preserves the tree-to-tree ordering of woody volume almost perfectly (R^2 = 0.971) while systematically reconstructing about half the destructive reference volume (bias −49.2%, median ratio 0.49) — consistent with the QSM representing only the woody skeleton and excluding fine branches and bark, whereas the destructive reference includes all above-ground woody

material. The volume offset is consistent with the QSM/allometry ratio in Fig. 2 (median 0.49 here, 0.64 against Chave).



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